COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	9/9/2026	**DATE SUBMITTED:**	9/10/2026
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ACTIVITY NO. __7_ Sort Techniques

1. PROCEDURE OUTPUT

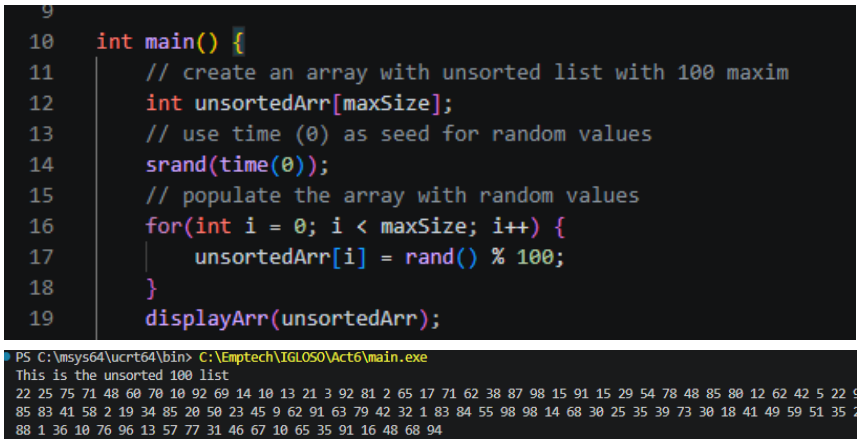
Code + Console Screenshot	
Observations	This part generates 100 random numbers using rand() and puts them in an array.

Table 7-1. Array of Values for Sort Algorithm Testing

Code + Console Screenshot	

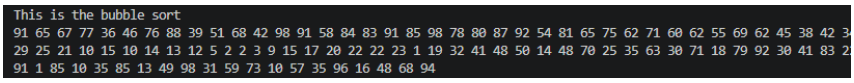
	
Observations	The code compares adjacent numbers and swap them if the left one is bigger than the right one. It keeps looping through the array until the highest numbers float to the end one by one.

Table 7-2. Bubble Sort Technique

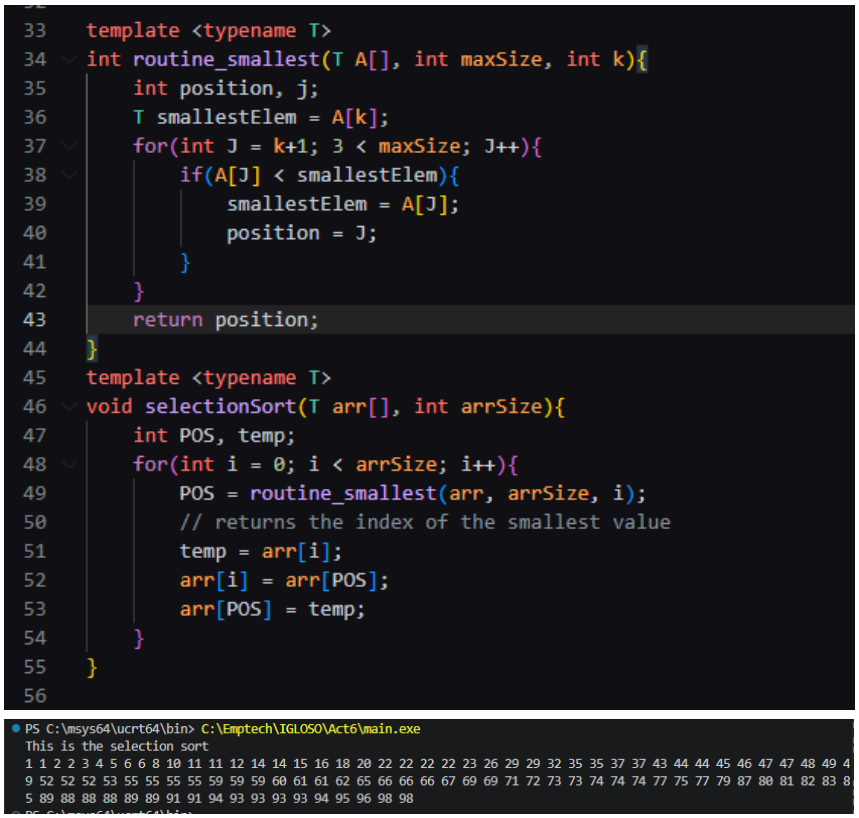
Code + Console Screenshot	
Observations	It uses the routine_smallest function to search for the smallest number in the unsorted portion of the array. Once found it swaps that element into its correct place starting from the front.

Table 7-3. Selection Sort Algorithm

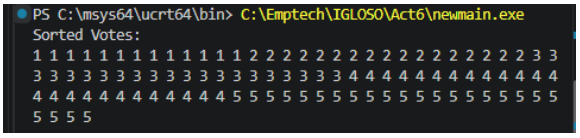
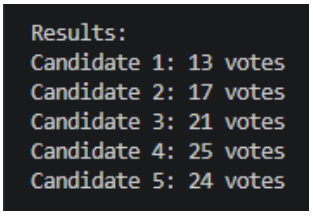
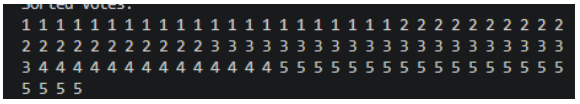
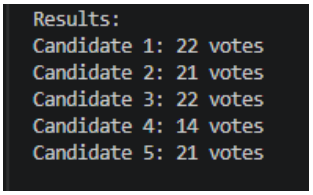
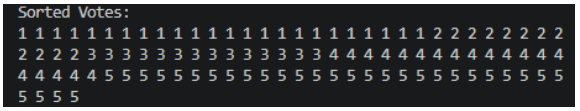
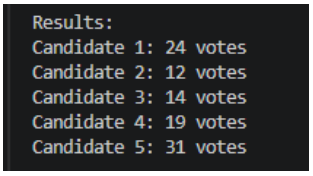
Code + Console Screenshot	
Observations	This algorithm works by taking one element at a time and shifting larger elements to the right to make space for it. It basically builds a sorted sub-list on the left side as it loops through.

```
17  template <typename T>
18  void insertionSort(T arr[], int arrSize){
19      int k = 1, j, temp;
20
21      while(k < arrSize){
22          temp = arr[k];
23          j = k-1;
24          while(j >= 0 && temp <= arr[j]){
25              arr[j+1] = arr[j];
26              j--;
27          }
28          arr[j+1] = temp;
29          k++;
30      }
31  }
```

```
PS C:\msys64\ucrt64\bin> C:\Emptech\IGLOSO\Act6\main.exe
This is the insertion sort
0 0 1 1 2 2 3 5 5 8 9 10 10 11 12 12 12 13 15 16 16 17 18 20 21 21 25 28 29 30 31 32 34 35 36 36 39 41 41 43 44 44 45
 46 46 46 48 48 48 49 51 51 52 52 53 54 54 55 56 56 57 57 59 59 62 63 65 65 65 66 69 71 72 75 75 76 76 76 78 83 83 83
 84 84 84 86 87 89 91 92 93 93 93 94 94 94 97 98 99 99
```

Table 7-4. Insertion Sort Algorithm

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Output Console Showing Sorted Array	Manual Count	Count Result of Algorithm
		
		
		

Question: Was your developed vote counting algorithm effective? Why or why not?

- Yes, it was effective since the sorting of the array helped make the data clean to process. This was due to the fact that the votes that were similar were arranged in a particular order by using the insertion sort method, making it easy for me to go through the array and sum the votes of each contestant.

CONCLUSION

- From this activity, I understood the principles of implementation of such algorithms as bubble sort, selection sort, and insertion sort in C++, and how important it is to sort the data before doing any calculations with it. Completing this activity, I observed the process of sorting according to each method and how the elements were moved by them; besides, I found out that one can very easily create an infinite loop due to a typing error in the condition of the loop. Concerning the additional activity, the use of insertion sort in relation to the voting list was very helpful for counting the number of votes as all the numbers of candidates appeared to be arranged properly. Generally, I managed to do the assignment well, and now I should continue improving my skills in debugging.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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